

SOP-01: Flammable and Combustible Organic Solvents

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Absolute ethanol / Molecular biology grade ethanol (BP2818)	64-17-5
Ethanol 190 proof	64-17-5
Decon's Ethanol, 200 proof	64-17-5
Ethanol for molecular biology (3916)	64-17-5
Ethyl alcohol, anhydrous, denatured (ex0285-3)	64-17-5 (denatured)
2-Propanol (isopropanol), BioReagent for molecular biology, >=99.5%	67-63-0
Acetonitrile	75-05-8

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Flame (flammable); Exclamation mark (irritant / acute toxicity Cat 4)

- Stock chemicals: neat ethanol (95-100%), denatured ethanol, 2-propanol and acetonitrile in glass or plastic bottles up to 4 L, and one 4 gal (approx. 15 L) drum of 190 proof ethanol. Intermediates and working solutions: 70%, 75% and 80% ethanol wash solutions prepared in nuclease-free water for nucleic acid precipitation; 100% isopropanol used neat for precipitation. Wastes: ethanol/isopropanol washes contaminated with guanidinium salts, phenol or

chaotropes from extraction kits; acetonitrile-containing rinses. No catalysts are used and no exothermic reactions are performed with these solvents in this lab.

- Primary physical hazard is flammability. Ethanol and 2-propanol are NFPA/IFC Flammable Liquid Class IB (flash point below 73 degF, boiling point at or above 100 degF) and GHS Flammable liquids Category 2. Acetonitrile is also Class IB / Category 2. Vapors are heavier than air, can travel to a distant ignition source and flash back, and form explosive mixtures with air. Static discharge during pouring or transfer of the 190 proof ethanol drum is a credible ignition source.
- Routes of exposure: inhalation of vapor, skin and eye contact, and accidental ingestion.
- How exposure might occur: aliquoting from bulk containers, open-bench decanting of ethanol washes during nucleic acid extraction, vortexing or centrifuging open tubes, spray-bottle use for bench and equipment decontamination, and splash during waste consolidation.
- Target organs: central nervous system, eyes, respiratory tract, skin, liver and kidney. Acetonitrile is additionally metabolized to cyanide and targets the cardiovascular and central nervous systems.
- Signs and symptoms of exposure: eye irritation, redness and tearing; dry, cracked or defatted skin; headache, dizziness, drowsiness, nausea and incoordination at high vapor concentrations. Acetonitrile overexposure may cause delayed onset (up to several hours) of weakness, chest tightness, rapid breathing, nausea, vomiting and convulsions; treat any suspected acetonitrile overexposure as a medical emergency even if the person initially feels well.
- GHS classification (from MyChem / SDS): Ethanol and 2-propanol - Flammable liquids Cat 2; Serious eye damage/eye irritation Cat 2 or 2A; 2-propanol also Specific target organ toxicity single exposure Cat 3 (drowsiness/dizziness). Acetonitrile - Flammable liquids Cat 2; Acute toxicity Cat 4 (oral, dermal, inhalation); Serious eye damage/eye irritation Cat 2.
- Occupational exposure limits (WISHA/OSHA PELs, 8-hr TWA): ethanol 1,000 ppm; 2-propanol 400 ppm; acetonitrile 40 ppm. These chemicals are NOT flagged as particularly hazardous substances in MyChem; PHS Sections 9-11 are therefore optional and are included below only as good practice.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- A certified chemical fume hood in FTR 209 is used for all transfers of acetonitrile, for any transfer of more than 500 mL of ethanol or 2-propanol, and for any operation in which solvent is heated or evaporated. Any chemical fume hood used must be tested and passed by EH&S; the current EH&S sticker must be present and the sash kept at or below the marked working height.
- Routine small-volume work (pipetting less than 50 mL of ethanol or isopropanol into microcentrifuge tubes during nucleic acid extraction) may be performed on the open bench because vapor generation is negligible and the room is mechanically ventilated.
- All ignition sources (open flames, Bunsen burners, hot plates, heat guns, non-explosion-proof heating blocks) are kept at least 3 m from open solvent containers. Bulk ethanol is dispensed only into UL-listed metal or approved plastic safety containers; the 190 proof drum is dispensed with the drum and receiving container bonded and grounded.
- Flammable liquids in excess of working quantities are stored in the flammable-liquid storage cabinet in FTR 209. Solvents are never stored in a standard (non-explosion-safe) laboratory refrigerator or freezer; only the flammable-safe or explosion-proof unit is used for chilled ethanol/isopropanol.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff. A Class B (or ABC) fire extinguisher and a solvent spill kit are maintained in the room.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: standard 100% cotton or cotton/poly barrier lab coat is acceptable for routine microliter-to-milliliter work. Because a risk of fire exists whenever more than 1 L of flammable solvent is handled at once, a flame-resistant, NFPA 2112-compliant lab coat must be worn when dispensing from the 4 gal ethanol drum or from any container larger than 1 L. FR coats are available from the SAFS lab coat service; note that some FR fabrics (Nomex, Rhovyl, Kevlar) are permeable and do not provide chemical/acid resistance.
- A chemically resistant apron and disposable sleeves are worn for bulk drum dispensing.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: disposable nitrile examination gloves, 4-8 mil (e.g., Kimberly-Clark KleenGuard G10 or Microflex MidKnight), for incidental contact with ethanol and 2-propanol. Nitrile provides only splash (incidental contact) protection against these alcohols and against acetonitrile - it is NOT suitable for immersion.
- For acetonitrile handling, or for any task where hands may be in prolonged contact with solvent, wear butyl rubber or laminate (Ansell Barrier / North Silver Shield) gloves over the nitrile inner glove. Acetonitrile breaks through nitrile in minutes.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes. Change nitrile gloves immediately after any solvent contact and at least every 30 minutes of continuous use.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 safety glasses with side shields for routine microliter-scale pipetting. Chemical splash goggles are required whenever pouring, decanting, or transferring volumes greater than 100 mL, and whenever handling acetonitrile. A face shield worn over goggles is required for drum dispensing and for spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP when they are performed as written.
- Ethanol and 2-propanol may be used outside of the fume hood in volumes up to 500 mL per operation on the open bench. Acetonitrile must NEVER be used outside of the certified chemical fume hood. Evaporation, heating, or sonication of any of these solvents must be performed in the fume hood.
- No lab member in this SOP is currently enrolled in the EH&S Respiratory Protection Program. If work is proposed that cannot be performed in the fume hood and could exceed an Action Limit or Occupational Exposure Limit, work must

stop and the PI must contact EH&S before proceeding; program enrollment includes medical evaluation, training and fit testing.

- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted). Where air-purifying respirators are appropriate, use a full-face respirator with organic vapor cartridges as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 2 L of flammable liquid may be outside the flammable storage cabinet at any one time, and no more than 500 mL may be open on the bench. Total flammable liquid storage in FTR 209 must remain within the maximum allowable quantity for the control area; do not order additional bulk solvent without checking cabinet capacity.
- Glass containers larger than 1 L must be transported in a rubber or polyethylene bottle carrier. The 190 proof ethanol drum is not moved between rooms; it is dispensed in place.
- Bonding and grounding: when dispensing from the 4 gal ethanol drum, connect the bonding strap between the drum and the receiving container before opening any valve. Refer to the EH&S Grounding and Bonding focus sheet.
- Ethanol spray bottles used for surface decontamination must be labeled with the full chemical name and hazard pictograms, kept below 500 mL, capped when not in use, and never sprayed near an open flame or an operating heating element.
- Storage: flammable-liquid storage cabinet, FTR 209. Store away from oxidizers (copper(II) nitrate, sulfuric acid, nitric acid), strong acids and strong bases. Acetonitrile is additionally kept segregated from strong acids and oxidizers. Keep all containers tightly closed and upright with secondary containment trays under bulk bottles.
- Do not store flammable solvents in a household-type refrigerator or freezer, on the floor, in a fume hood as long-term storage, or above eye level.
- Unattended operations involving these solvents (e.g., overnight evaporation) are prohibited without written PI approval and a posted unattended-operation sign giving the chemical, the responsible person and a contact number.
- Temporary hazard signage is posted on the fume hood sash when acetonitrile work is in progress.
- Clean the chemical fume hood work surface upon completion of tasks with 70% ethanol followed by deionized water; wipe dry.
- Clean all contaminated bench surfaces with 70% ethanol and dry. Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these solvents must be decontaminated at the end of the laboratory work session; allow residual solvent to evaporate inside the fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Immediately on discovering a spill of any size: eliminate all ignition sources in the area, alert others in the room, and do not switch electrical equipment on or off.

- Small spill - less than 500 mL inside the fume hood, or less than 100 mL on the open bench: may be handled by trained lab staff. Wearing safety glasses or goggles, nitrile gloves (butyl over-gloves for acetonitrile) and a lab coat, cover the spill with the universal absorbent pads or vermiculite from the laboratory solvent spill kit, work from the outside of the spill inward, and place the used absorbent into a sealed, labeled polyethylene bag or screw-top container.
- Large spill - more than 500 mL inside the hood, more than 100 mL on the open bench, ANY quantity of acetonitrile outside the fume hood, or any spill from the 4 gal ethanol drum: DO NOT clean up. Evacuate the room, close the door, post a warning, and call 9-1-1 from a campus phone. Notify EH&S.
- Liquid spills only are anticipated for this SOP; there is no powder form of these chemicals in this inventory.
- PPE required for cleanup: safety goggles, face shield for volumes greater than 500 mL, nitrile gloves (plus butyl over-gloves for acetonitrile), buttoned lab coat (FR lab coat for drum spills), full-length pants and close-toe shoes.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the spill area thoroughly with a mild detergent solution followed by clean water, then dry.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables in plastic bags, place in a screw-top container, label with a UW Hazardous Waste Label listing the solvent(s) and 'spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Inhalation exposure: move the person out of the contaminated area to fresh air; get medical help. For acetonitrile, call 9-1-1 immediately even if the person appears well, because symptoms of cyanide toxicity can be delayed several hours.
- Skin exposure: use the nearest safety shower for 15 minutes; stay under the shower and remove contaminated clothing; use a clean lab coat or spare clothing for cover-up.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open.
- Ingestion: do not induce vomiting; rinse mouth with water and call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of the exposure: agent, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams generated under this SOP: (a) neat or dilute ethanol and 2-propanol washes free of other hazardous constituents; (b) alcohol washes contaminated with guanidinium thiocyanate, phenol or other extraction reagents; (c) acetonitrile-containing waste; (d) solvent-contaminated solid debris (pipette tips, wipes, absorbent).
- Keep these streams SEPARATE. In particular, never combine alcohol waste with nitric acid, sulfuric acid, or other oxidizer waste; never combine acetonitrile waste with acid waste (hydrogen cyanide evolution).

- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible container (HDPE or the original solvent bottle) with a securely closable screw-top lid. Keep the container closed except when actively adding waste. Use secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Do NOT pour ethanol, 2-propanol or acetonitrile down the drain. Do NOT evaporate solvent waste in the fume hood as a disposal method.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- Solid debris lightly contaminated with ethanol (pipette tips, gloves, wipes) may be allowed to air-dry in the fume hood and then be discarded as regular laboratory solid waste. Debris saturated with solvent, or contaminated with acetonitrile, is collected as hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Decontamination of equipment, glassware and work areas: rinse glassware three times with deionized water, collecting the first rinse as hazardous waste if acetonitrile was used; wipe the fume hood surface and bench with 70% ethanol and then deionized water. Update the MyChem inventory when a container is emptied or a new container is received.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplus materials.

Section 7 - Protocol

- Procedure A - Ethanol precipitation and washing of nucleic acids (open bench, microliter scale). Add 2-2.5 volumes of chilled 100% ethanol (or 0.7 volume 100% isopropanol) plus 0.1 volume 3M sodium acetate pH 5.2 to the aqueous nucleic acid sample in a 1.5 mL microcentrifuge tube; cap, invert to mix, and incubate at -20 degC. Centrifuge, decant supernatant into the labeled alcohol waste container, wash the pellet twice with 70-80% ethanol, decant to waste, air-dry the pellet at room temperature with the tube open on the bench, and resuspend in nuclease-free water or 1X TE. All tubes are capped before vortexing or centrifuging.
- Procedure B - Preparation of working ethanol solutions. In the fume hood, dilute 100% or 190 proof ethanol with nuclease-free or deionized water to 70%, 75% or 80% in a labeled bottle. Label with the full chemical name, concentration, date and preparer's initials. Do not use unlabeled squeeze bottles.
- Procedure C - Surface decontamination. Apply 70% ethanol from a labeled spray bottle to benches, pipettes and equipment; wipe with a lint-free wipe. Confirm no heating element, flame or hot surface is within 3 m before spraying.
- Procedure D - Acetonitrile handling (fume hood only). Perform all measuring, transfer and mixing of acetonitrile with the sash at the marked working height. Keep the container closed when not actively dispensing. Never acidify acetonitrile waste.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use.
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - none of the chemicals covered by this SOP are administered to animals. Ethanol used for euthanasia, anesthesia or fixation of live animals is outside the scope of this SOP and would require a separate IACUC-approved protocol.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with the solvents covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each solvent, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training: EH&S Laboratory Safety Training (initial and required refreshers) and EH&S Managing Laboratory Chemical Waste training must be completed and documented before independent work.
- A worker must have documented hands-on training from the PI or a designated senior lab member, including one supervised run of Procedures A-C, before performing them independently for the first time.
- Bulk dispensing from the 4 gal ethanol drum requires separate documented training on bonding and grounding and specific authorization from the PI.
- No medical examination or respirator authorization is required for the work described, because respirators are not used. If respirator use ever becomes necessary, a medical evaluation and fit testing through the EH&S Respiratory Protection Program must be completed first.

Section 10 - Decontamination

- Equipment: wipe pipettes, tube racks, centrifuge rotors and buckets with 70% ethanol and allow to air dry.
- Glassware: allow residual solvent to evaporate in the fume hood, then rinse three times with tap water and three times with deionized water. Glassware used for acetonitrile is rinsed first with water into the hazardous waste container before routine washing.
- Fume hood: wipe the work surface and interior sash with 70% ethanol followed by deionized water at the end of each work session.
- Bench and controlled areas: wipe the designated solvent bench area with 70% ethanol and dry; there are no glove boxes or restricted-access hoods in FTR 209.

Section 11 - Designated Area

- Ethanol and 2-propanol at the microliter-to-milliliter scale: the designated molecular biology bench in FTR 209 and the certified chemical fume hood in FTR 209.
- Acetonitrile, all bulk transfers (more than 500 mL), and all heating or evaporation of solvents: the certified chemical fume hood in FTR 209 only.
- Bulk storage: the flammable-liquid storage cabinet in FTR 209.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.

